We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

### Product name:
Palit

Use: crop protection product, growth regulator

Manufacturer/supplier:
BASF Vietnam Co. Ltd.
Level 23, Deutsches Haus, 33 Le Duan,
Ben Nghe Ward, Dist. 1, Ho Chi Minh City, Vietnam
Telephone: +84 28 3824 3833
Telefax number: +84 28 3824 3832
E-mail address: minh-triet.thieu@basf.com

Emergency information:
18001703 (Vietnam)
Telefax number: +84 28 3824 3832
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Acute toxicity: Cat.3 (oral)
Acute toxicity: Cat.4 (Inhalation - mist)
Specific target organ toxicity — single exposure (Central nervous system): Cat.2
Specific target organ toxicity — repeated exposure (Central nervous system): Cat.2
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H332 | Harmful if inhaled. |
| H301 | Toxic if swallowed. |
| H371 | May cause damage to organs (Central nervous system). |
| H373 | May cause damage to organs (Central nervous system) through prolonged or repeated exposure. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P271 | Use only outdoors or in a well-ventilated area. |
| P260 | Do not breathe dust/gas/mist/vapours. |
| P264 | Wash contaminated body parts thoroughly after handling. |
| P270 | Do not eat, drink or smoke when using this product. |

Precautionary Statements (Response):

| | |
|---|---|
| P308 + P311 | IF exposed or concerned: Call a POISON CENTER or physician. |
| P301 + P310 | IF SWALLOWED: Immediately call a POISON CENTER or physician. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P314 | Get medical advice/attention if you feel unwell. |
| P330 | Rinse mouth. |
| P391 | Collect spillage. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

May produce an allergic reaction. Contains:
mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, insecticide, suspension concentrate (SC)

**Hazardous ingredients**

Chlorfenapyr

| | |
|---|---|
| Content (W/W): 10 % | Acute Tox.: Cat. 2 (Inhalation - dust) |
| CAS Number: 122453-73-0 | Acute Tox.: Cat. 3 (oral) |
| | STOT SE (Central nervous system): Cat. 2 |
| | STOT RE (Central nervous system): Cat. 2 |
| | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 100 |

Residues (petroleum), catalytic reformer fractionator, sulfonated,polymers with formaldehyde, sodium salts

| | |
|---|---|
| Content (W/W): < 5 % | Eye Dam./Irrit.: Cat. 2A |
| CAS Number: 68425-94-5 | Aquatic Acute: Cat. 3 |
| | Aquatic Chronic: Cat. 3 |

mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

| | |
|---|---|
| Content (W/W): < 0.01 % | Acute Tox.: Cat. 3 (oral) |
| CAS Number: 55965-84-9 | Acute Tox.: Cat. 2 (Inhalation - mist) |
| | Acute Tox.: Cat. 2 (dermal) |
| | Skin Corr.: Cat. 1C |
| | Eye Dam.: Cat. 1 |
| | Skin Sens.: Cat. 1A |
| | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 100 |

propane-1,2-diol

Content (W/W): < 10 %
CAS Number: 57-55-6

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200 - 300 ml water, do not induce vomiting, seek medical attention. If swallowed, seek medical advice immediately and show this container or label. Do not induce vomiting unless told to by a poison control center or doctor. Take patient to hospital immediately. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product. Medical monitoring for at least 7 days.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Symptoms of poisoning may only appear after several hours or several days.
Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen bromide, hydrogen chloride, hydrogen fluoride, nitrogen oxides, sulfur oxides, silica compounds, halogenated compounds, aluminum compounds
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

## 6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray.Use personal protective clothing.Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into drains/surface waters/groundwater.Do not discharge into the subsoil/soil.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 60 Months

Protect from temperatures below: 0 °C
The product can crystallize below the limit temperature.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

Chlorfenapyr, 122453-73-0;

TWA value 0.3 mg/m3 (BASF recomm. occupational exposure limit)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid |
| Colour: | off-white |
| Odour: | faint odour, bitter almond-like |
| Odour threshold: | Not determined since harmful by inhalation. |
| pH value: | approx. 6 - 8<br>(1 %(m), 21 °C) |
| Freezing point: | approx. 0 °C<br>Information applies to the solvent. |
| Boiling point: | approx. 100 °C<br>Information applies to the solvent. |
| Flash point: | Non-flammable. |
| Evaporation rate: | not applicable |
| Flammability (solid/gas): | not applicable |

| | | |
|---|---|---|
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. | |
| Ignition temperature: | approx. 520 °C | (Directive 92/69/EEC, A.15) |
| Thermal decomposition: | 252 °C , 240 kJ/kg | (DSC (OECD 113)) |
| | 310 °C , 130 kJ/kg | (DSC (OECD 113)) |
| | 450 °C , > 90 kJ/kg | (DSC (OECD 113)) |
| | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. | |
| SADT: | > 75 °C<br>Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4) | |
| Explosion hazard: | not explosive | |
| Fire promoting properties: | not fire-propagating | (Directive 2004/73/EC, A.21) |
| Vapour pressure: | approx. 23.3 hPa<br>(20 °C)<br>Information applies to the solvent. | |
| Density: | approx. 1.07 g/cm3<br>(20 °C) | (OECD Guideline 109) |
| Relative vapour density (air): | not applicable | |
| Solubility in water: | dispersible | |
| Partitioning coefficient n-octanol/water (log Pow): | not applicable for mixtures | |
| Viscosity, dynamic: | approx. 68 mPa.s<br>(20 °C, 100 1/s) | (OECD Guideline 114) |

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics

| | |
|---|---|
| Particle size distribution: | The substance / product is marketed or used in a non solid or granular form. - |

## 10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

| | |
|---|---|
| Thermal decomposition: | 252 °C, 240 kJ/kg (DSC (OECD 113)) |
| Thermal decomposition: | 310 °C, 130 kJ/kg (DSC (OECD 113)) |
| Thermal decomposition: | 450 °C, > 90 kJ/kg (DSC (OECD 113)) |
| Thermal decomposition: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

**Acute oral toxicity**

Experimental/calculated data:
LD50mouse (oral): > 50 - < 300 mg/kg
The data on toxicology refer to the active ingredient. Same product LD50 is assigned for precautionary reasons in view of human poisoning incidents

**Acute oral toxicity**

LD50rat (oral): > 500 - < 2,000 mg/kg (OECD Guideline 423)

**Acute inhalation toxicity**

LC50 rat (by inhalation): 2.62 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

**Acute dermal toxicity**

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)
No mortality was observed.

**Assessment of acute toxicity**

Of pronounced toxicity after single ingestion. Of moderate toxicity after short-term inhalation. Virtually nontoxic after a single skin contact.

Information on: Chlorfenapyr

**Acute oral toxicity**

Experimental/calculated data:
LD50 mouse (oral): > 50 - < 300 mg/kg (OECD Guideline 401)
---

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far
Symptoms of poisoning may only appear after several hours or several days.

## Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

## Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
Mouse Local Lymph Node Assay (LLNA) mouse: Non-sensitizing.

## Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

## Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

## Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

## Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

### Specific target organ toxicity (single exposure)

A single exposure may have relevant toxic effects on organs.

Target organ: Central nervous system
Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Affects the central nervous system.

Information on: mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)
Assessment of repeated dose toxicity:
Based on available data, the classification criteria are not met. After repeated exposure the prominent effect is local irritation.
---

### Aspiration hazard

not applicable

### Other relevant toxicity information

Reported human health effects after incorporation: immediate symptoms after oral ingestion may comprise nausea, vomiting, sweating and abdominal pain. Subsequent symptoms emerging with a delay of a few days up to 10-14 days may include drowsiness, tachypnea, tachycardia or arrhythmia, hypertension, fever, diaphoresis, muscle fasciculation, muscle stiffness, weakness of limbs (sometimes leading to paralysis), hepatic and renal dysfunction, pancreatitis, and sudden disturbance of consciousness followed by coma and cardiac arrest. Misuse can be fatal.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) 0.463 mg/l, Cyprinus carpio

Aquatic invertebrates:
EC50 (48 h) 0.0407 mg/l, Daphnia magna

Aquatic plants:
EC50 (72 h) 126 mg/l (growth rate), Pseudokirchneriella subcapitata

No observed effect concentration (72 h) 0.772 mg/l (growth rate), Pseudokirchneriella subcapitata

Information on: Chlorfenapyr
Chronic toxicity to fish:
No observed effect concentration (93 d) 0.003678 mg/l, Oncorhynchus mykiss
---

Information on: Chlorfenapyr
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (28 d), 0.000172 mg/l, Mysidopsis bahia
---

## Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.
---

## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).
---

## Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Bioaccumulation potential:
Bioconcentration factor: 116, Cyprinus carpio
Accumulation in organisms is not to be expected.
---

### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Information on disposal regulations: Must comply with regulations on hazardous waste such as:

- Decree No. 08/2022/ND-CP of the Government: detailing a number of articles of the Law on Environmental Protection.

- Circular No. 02/2022/TT-BTNMT of the Ministry of Natural Resources and Environment: detailing a number of articles of the Law on Environmental Protection.

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 2902 |
| UN proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR) |
| Transport hazard class(es): | 6.1, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 2902 |
| UN proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR) |
| Transport hazard class(es): | 6.1, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-A; S-A |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 2902 |
| UN proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR) |

| | |
|---|---|
| Transport hazard class(es): | 6.1 |
| Packing group: | III |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

**Other regulations**

Legal information on safety, health and environment for chemicals:
Law on Chemicals No. 06/2007/QH12
Chemical Safety Sheets (SDS) are compiled in accordance with international regulations on chemical classification and labelling (GHS) as well as relevant applicable legal regulations:
- QCVN 05A:2020/BCT National Technical Regulation on safety in production, trading, use, preservation and transportation of hazardous chemicals.
- Amendment 1:2024 QCVN 05A:2020/BCT National technical regulation on safety in production, trading, use, preservation and transportation of hazardous chemicals.
- Decree No. 113/2017/ND-CP dated 09/10/2017 of the Government detailing and guiding the implementation of a number of articles of the Law on Chemicals;
- Circular No. 32/2017/TT-BCT dated 28/12/2017 of the Ministry of Industry and Trade specifying and guiding the implementation of a number of articles of the Law on Chemicals and Decree No. 113/2017/ND-CP dated 09/10/2017 of the Government detailing and guiding the implementation of a number of articles of the Law on Chemicals;
- Decree No. 82/2022/ND-CP dated October 18, 2022 of the Government amending and supplementing a number of articles of the Government's Decree No. 113/2017/ND-CP dated October 09, 2017 detailing and guiding the implementation of a number of articles of the Law on Chemicals
- Circular No. 17/2022/TT-BCT dated October 27, 2022 of the Ministry of Industry and Trade amending and supplementing a number of articles of Circular No. 32/2017/TT-BCT dated December 28, 2017 of the Minister of Industry and Trade specifying and guiding the implementation of a number of articles of the Law on Chemicals and the Government's Decree No. 113/2017/ND-CP dated October 09, 2017 detailing and guiding the exam a number of articles of the Law on Chemicals
- Decree No. 161/2024 ND-CP dated December 18, 2024 of the Government regulating the list of dangerous goods, transportation of dangerous goods and the order and procedures for issuance of licenses and certificates of completion of training programs for drivers or escorts transporting dangerous goods on roads;
- Decree 34/2024/ND-CP dated 31/03/2024 regulating the list of dangerous goods, transporting dangerous goods by road motor vehicles and inland waterway vehicles
- Circular No. 37/2020/TT-BCT dated 30/11/2020 of the Ministry of Industry and Trade stipulating the list of dangerous goods that must be packed in the process of transporting and transporting dangerous goods by road, railway and inland waterway motor vehicles.
- Decree No. 71/2019/ND-CP dated August 30, 2019 of the Government stipulating the sanctioning of administrative violations in the field of chemicals and industrial explosives.

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.